

Powerful values of $2z^3 + 1$

Elementary reductions and conditional finiteness

The integer equation $x^3y^2 = 2z^3 + 1$

Scope. An unconditional finiteness proof is not established in this note. What follows is a self-contained reduction of the question, elementary necessary conditions, and a complete finiteness proof *assuming the abc conjecture*. That conditional theorem does not answer the requested unconditional claim. No internet sources were consulted; no claim of novelty or of a complete solution list is made.

1. The exact arithmetic issue

A positive integer n is called *powerful* if every prime dividing n has exponent at least two in its prime factorization. The integer 1 is powerful by this definition. Since $2z^3 + 1$ is odd, it is never zero for $z \in \mathbb{Z}$.

Proposition 1. *For a fixed integer z , there exist integers x, y satisfying*

$$x^3y^2 = 2z^3 + 1 \tag{1}$$

if and only if $|2z^3 + 1|$ is powerful. Each such z gives only finitely many pairs (x, y) .

Proof. Put $n = |2z^3 + 1|$. If (1) holds, then $x, y \neq 0$, and the exponent of a prime p in n is $3\alpha + 2\beta$, where $\alpha, \beta \geq 0$ are its exponents in $|x|, |y|$. A positive integer of that form is at least two. Thus n is powerful.

Conversely, write $n = \prod_p p^{e_p}$, where every displayed $e_p \geq 2$. Set $\delta_p = 0$ when e_p is even and $\delta_p = 1$ when it is odd. In the odd case $e_p \geq 3$, so the integers

$$a = \prod_p p^{\delta_p}, \quad b = \prod_p p^{(e_p - 3\delta_p)/2}$$

are well defined and satisfy $n = a^3b^2$. Taking $x = \text{sgn}(2z^3 + 1)a$ and $y = \pm b$ proves existence. For $n = 1$ the products are empty and equal to 1. Finally, any solution has $|x|^3 \leq n$ and $|y|^2 \leq n$, so there are only finitely many pairs. $\square$

The construction makes $|x|$ squarefree. It is the unique representation with $|x|$ squarefree and $|y| > 0$, apart from the sign choice for y : reducing $e_p = 3\alpha + 2\beta$ modulo 2 forces $\alpha = \delta_p$ whenever $\alpha \in \{0, 1\}$.

Consequently, the original finiteness question is *exactly* the question whether $|2z^3 + 1|$ is powerful for only finitely many integers z . This is a reformulation of the obstacle, not its resolution.

2. Unconditional restrictions and verified solutions

Proposition 2. *Every integer solution of (1) satisfies*

$$x, y \text{ odd}, \quad \gcd(|x|, |z|) = \gcd(|y|, |z|) = 1,$$

and

$$z \not\equiv 1 \pmod{3}, \quad y \equiv \pm 1 \pmod{9}.$$

Also, $x > 0$ when $z \geq 0$, while $x < 0$ when $z \leq -1$.

Proof. The right side of (1) is odd, so both x and y are odd. If a prime divides z and either x or y , reducing the equation modulo that prime gives $0 = 1$, a contradiction. The sign assertion follows because $y^2 > 0$ and $2z^3 + 1$ is positive exactly when $z \geq 0$.

The cubes modulo 9 are 0, 1, -1, so the possible residues of the right side are 1, 3, -1. None is 0. Hence neither x nor y is divisible by 3. The possible left-side residues are therefore

$$(\pm 1)\{1, 4, 7\} = \{1, 2, 4, 5, 7, 8\},$$

which exclude 3. This rules out $z \equiv 1 \pmod{3}$. When $z \equiv 0 \pmod{3}$, the right side is 1 modulo 9; this forces $x^3 \equiv 1$ and $y^2 \equiv 1$. When $z \equiv -1 \pmod{3}$, the right side is -1; this forces $x^3 \equiv -1$ and again $y^2 \equiv 1$. Checking the nine residues gives $y^2 \equiv 1 \pmod{9}$ exactly for $y \equiv \pm 1 \pmod{9}$, as asserted. $\square$

The two values $z = 0, -1$ are completely elementary. At $z = 0$ the equation is $x^3y^2 = 1$, and at $z = -1$ it is $x^3y^2 = -1$. Taking absolute values forces $|x| = |y| = 1$. The resulting solutions are exactly

$$(1, 1, 0), (1, -1, 0), (-1, 1, -1), (-1, -1, -1)$$

within those two slices. No assertion is made that these are all integer solutions.

An unconditional obstruction to polynomial families

Proposition 3. *There are no polynomials $X, Y, Z \in \mathbb{Q}[T]$ with Z nonconstant satisfying $X^3Y^2 = 2Z^3 + 1$ identically.*

Proof. Suppose such an identity exists. Write $a = \deg X$, $b = \deg Y$, and $c = \deg Z > 0$. The polynomials X, Y are nonzero, and comparing degrees shows $3a + 2b = 3c$. Also $\gcd(XY, Z) = 1$, since any common polynomial divisor would divide the constant 1 in the identity.

Differentiation gives the nonzero polynomial

$$D = X^2Y(3X'Y + 2XY') = 6Z^2Z'.$$

The coprime polynomials X^2Y and Z^2 both divide D , so their product divides D . Taking degrees yields

$$2a + b + 2c \leq 3c - 1, \quad 2a + b \leq c - 1.$$

But $c = a + 2b/3$, so this would give $a + b/3 \leq -1$, contradicting $a, b \geq 0$. $\square$

This rules out a polynomial construction with varying z ; it does not rule out infinitely many isolated integer solutions. Likewise, finiteness for each fixed x would leave an infinite union of possible slices. An unconditional bound on z is still missing.

3. A complete conditional finiteness proof

For a positive integer m , let $\text{rad}(m)$ be the product of its distinct prime divisors, with $\text{rad}(1) = 1$.

The abc conjecture. *For every $\varepsilon > 0$ there is a constant $K_\varepsilon \geq 1$ such that all pairwise coprime positive integers A, B, C with $A + B = C$ satisfy*

$$C \leq K_\varepsilon \text{rad}(ABC)^{1+\varepsilon}.$$

Theorem 1 (Conditional on abc). *Assume the abc conjecture, and put $K = K_{1/11}$. Every integer solution of (1) with $z \neq 0$ satisfies*

$$\max\{|2z^3 + 1|, 2|z|^3\} \leq 256K^{11}.$$

In particular, under this assumption there are only finitely many integer solutions.

Proof. Let $t = |z| \geq 1$, $N = |2z^3 + 1|$ and $H = \max(N, 2t^3)$. For $z > 0$ use

$$(A, B, C) = (1, 2t^3, N),$$

and for $z < 0$ use $(A, B, C) = (1, N, 2t^3)$. In both cases $A + B = C = H$. The two numbers N and $2t^3$ differ by 1, so the triple is pairwise coprime, as required.

Proposition 1 implies $\text{rad}(N) \leq N^{1/2}$: squaring the product of its distinct primes gives a divisor of N . Also, $\text{rad}(2t^3) \leq 2t$. Since $N \leq H$ and $t \leq (H/2)^{1/3}$, we obtain

$$\text{rad}(ABC) \leq 2t\sqrt{N} \leq 2^{2/3}H^{5/6}. \quad (2)$$

Applying abc with $\varepsilon = 1/11$ now gives

$$H \leq K(2^{2/3}H^{5/6})^{12/11} = K2^{8/11}H^{10/11}.$$

Divide by $H^{10/11} > 0$ and raise to the eleventh power to obtain $H \leq 2^8K^{11}$. Thus $|z|^3 \leq 128K^{11}$, so only finitely many z are possible. Each admits finitely many pairs (x, y) by Proposition 1. The two solutions at $z = 0$ were already handled separately. $\square$

Intuition. Powerful numbers have relatively few distinct prime factors: their radical is at most their square root. In the consecutive pair $2t^3, N$, the cubic term contributes at most about $H^{1/3}$ to the radical and the powerful term at most $H^{1/2}$. Their product is bounded by a constant times $H^{5/6}$. The exponent $5/6 < 1$ leaves a gap that abc converts into a height bound. Without that conjecture, this radical estimate alone supplies no such bound.

Final logical status. The elementary reductions are unconditional; the finiteness theorem is conditional. No numerical value of K has been supplied, and neither an unconditional finiteness proof nor a complete enumeration is established here.